

Learning to Refer: How Scene Complexity Affects Emergent Communication in Neural Agents

Dominik Künkele¹ and Simon Dobnik^{1,2}

¹Department of Philosophy, Linguistics and Theory of Science

²Centre for Linguistic Theory and Studies in Probability (CLASP)

University of Gothenburg, Sweden

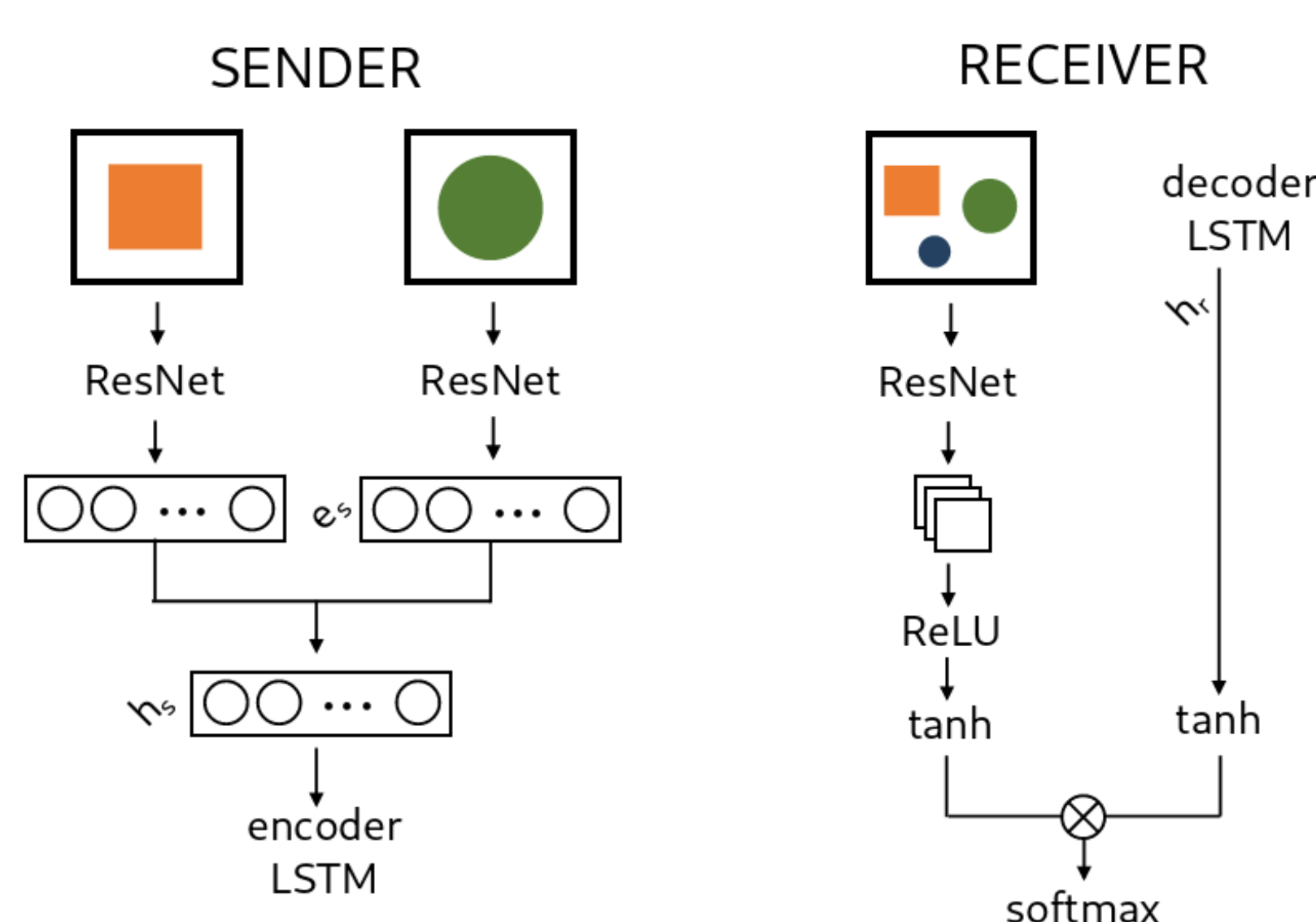
contact@dominik-kuenkele.de simon.dobnik@gu.se

Aims

A picture can say thousand words...

- Study **emergence of discrete linguistic symbols and languages** from continuous sensory input in **interactive communication**
- Study **constraints** on symbol grounding in interacting neural agents
- Investigate the **role of scene complexity, discriminating features present, and message space** on emergent languages

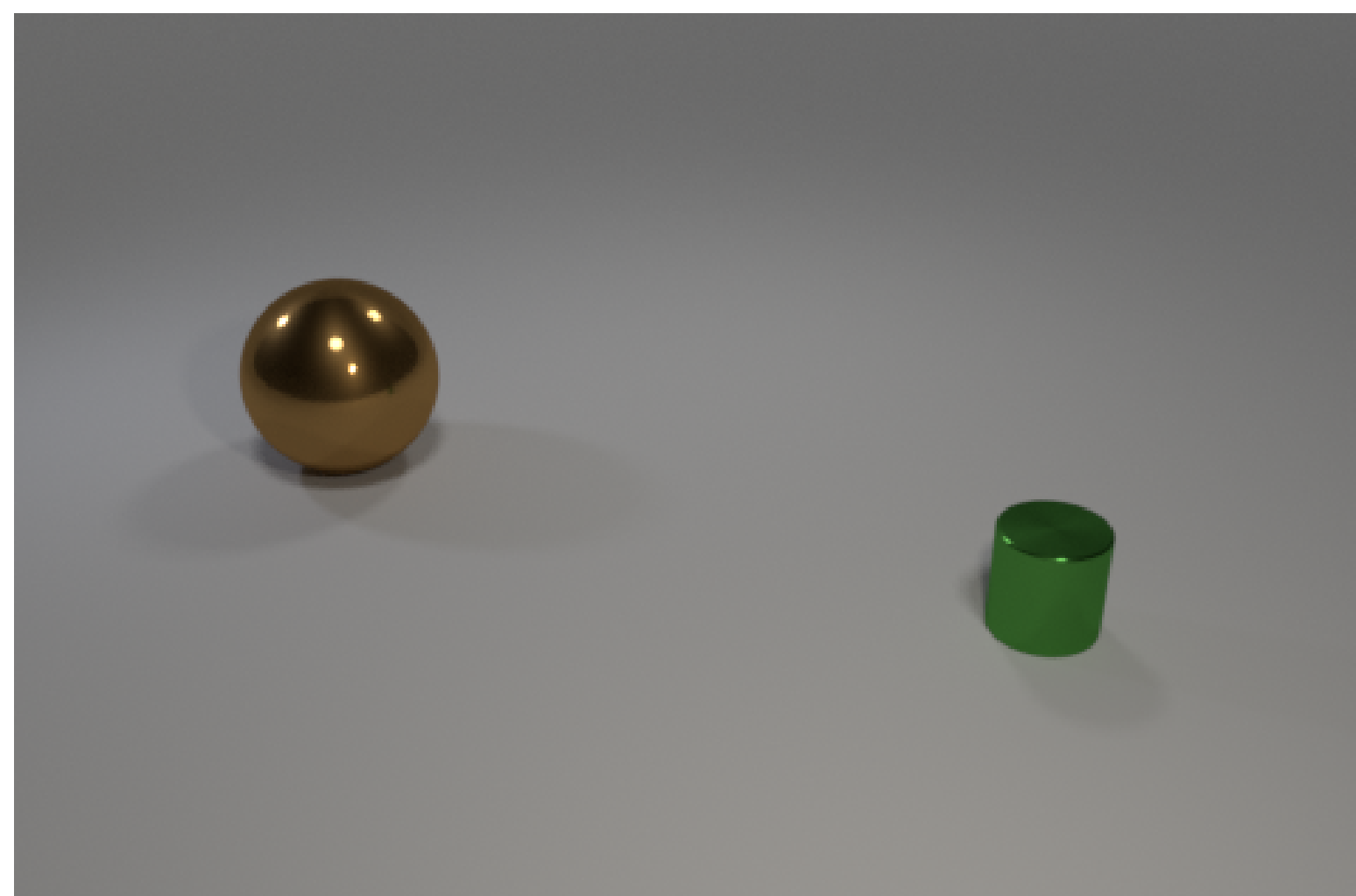
Architecture



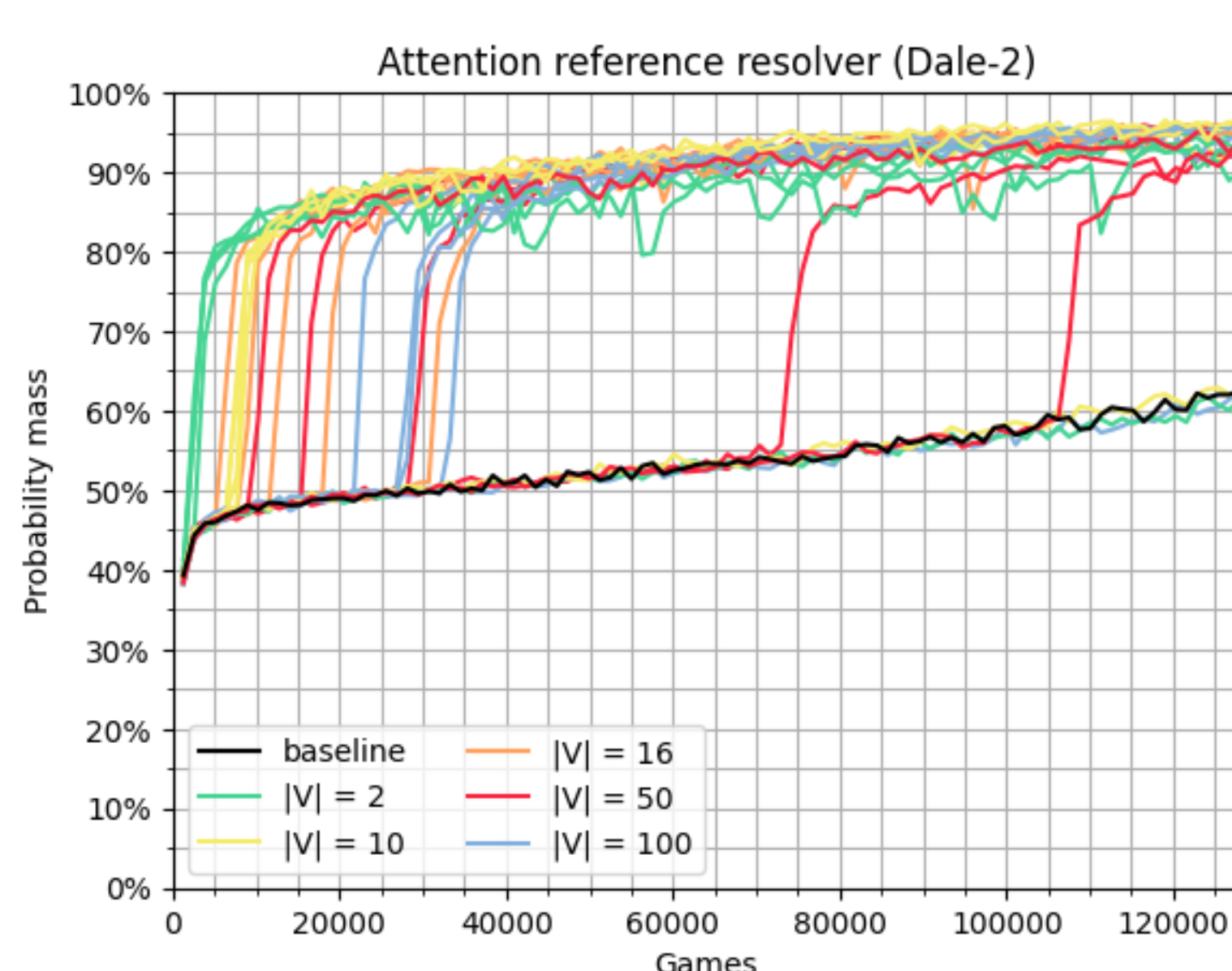
Conclusions

- Medium vocabularies ($|V| \in \{10, 16\}$) and message lengths ($n \in \{2, 4\}$) perform best
- Scene complexity (number of distractors, number of features) dramatically impacts communication success
- Shape attributes easiest to learn, size attributes most challenging
- Future: Analyze linguistic properties of emergent languages

Dale-2 Dataset



Target + 1 distractor

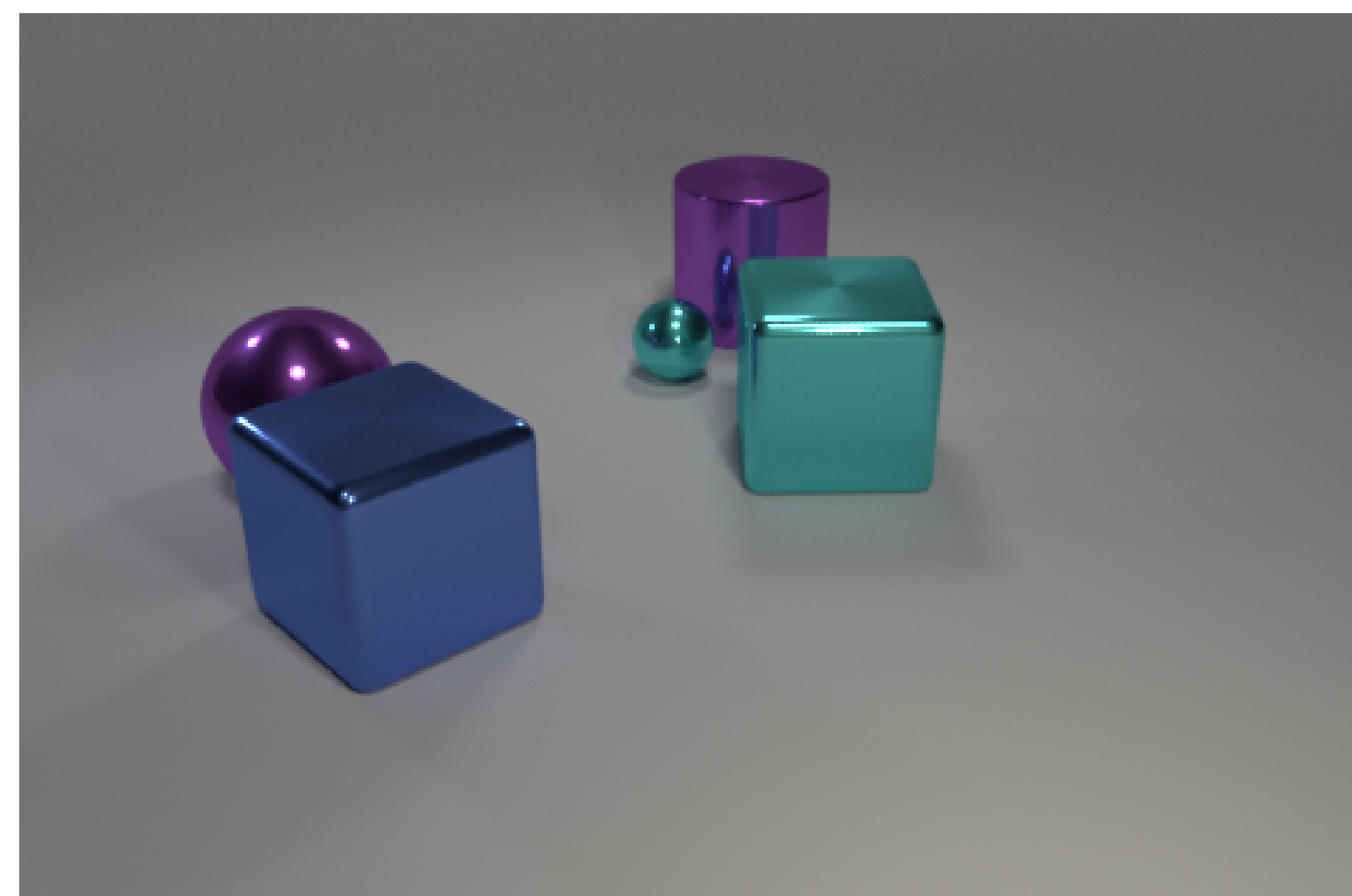


Learning curves by vocabulary size

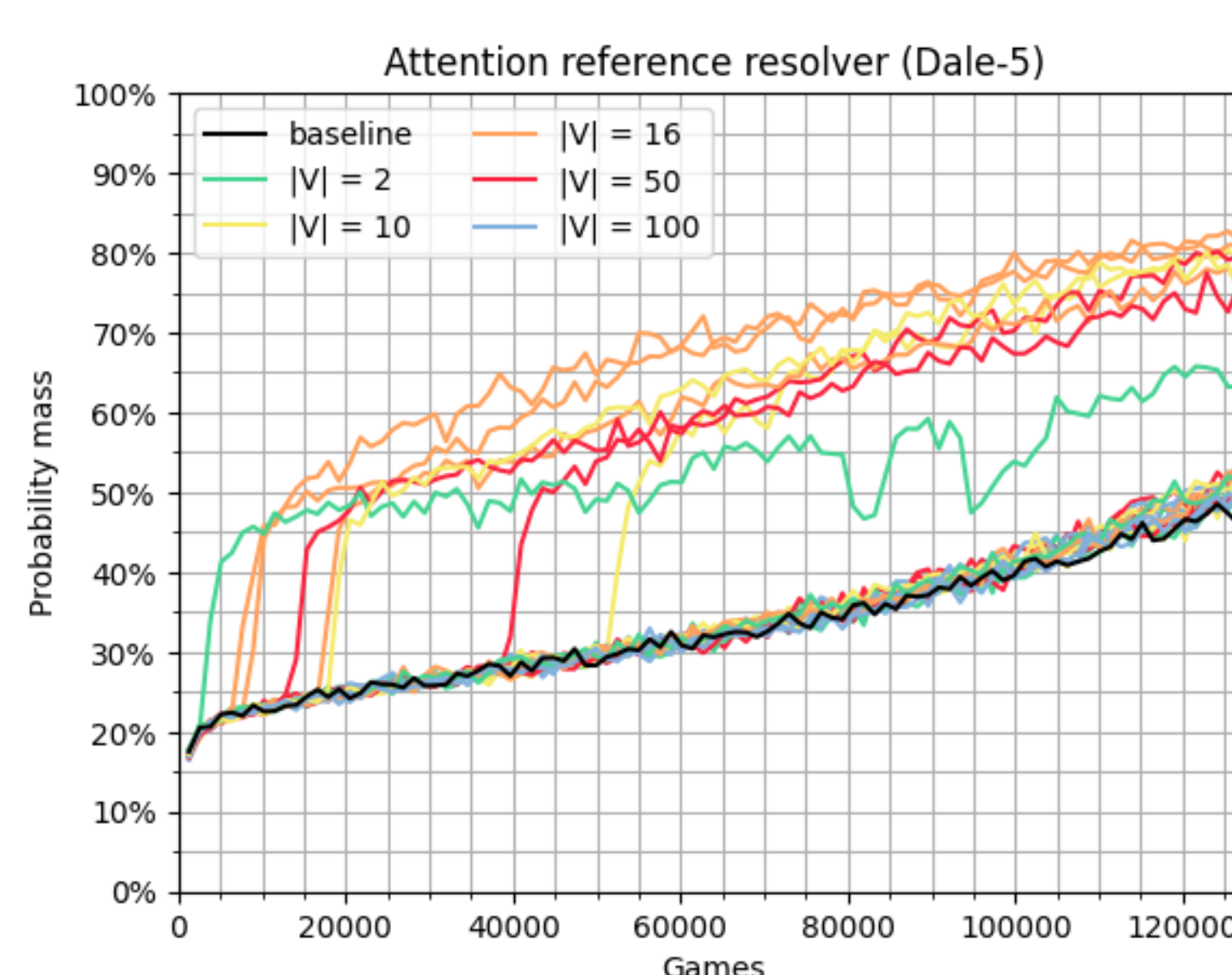
RE Generation Results:

- **Accuracy:** 99%
- **F1-Score:** 98.57%
- **Shape:** 99.75% precision
- **Color:** 98.09% precision
- **Size:** 98.73% precision

Dale-5 Dataset



Target + 4 distractors

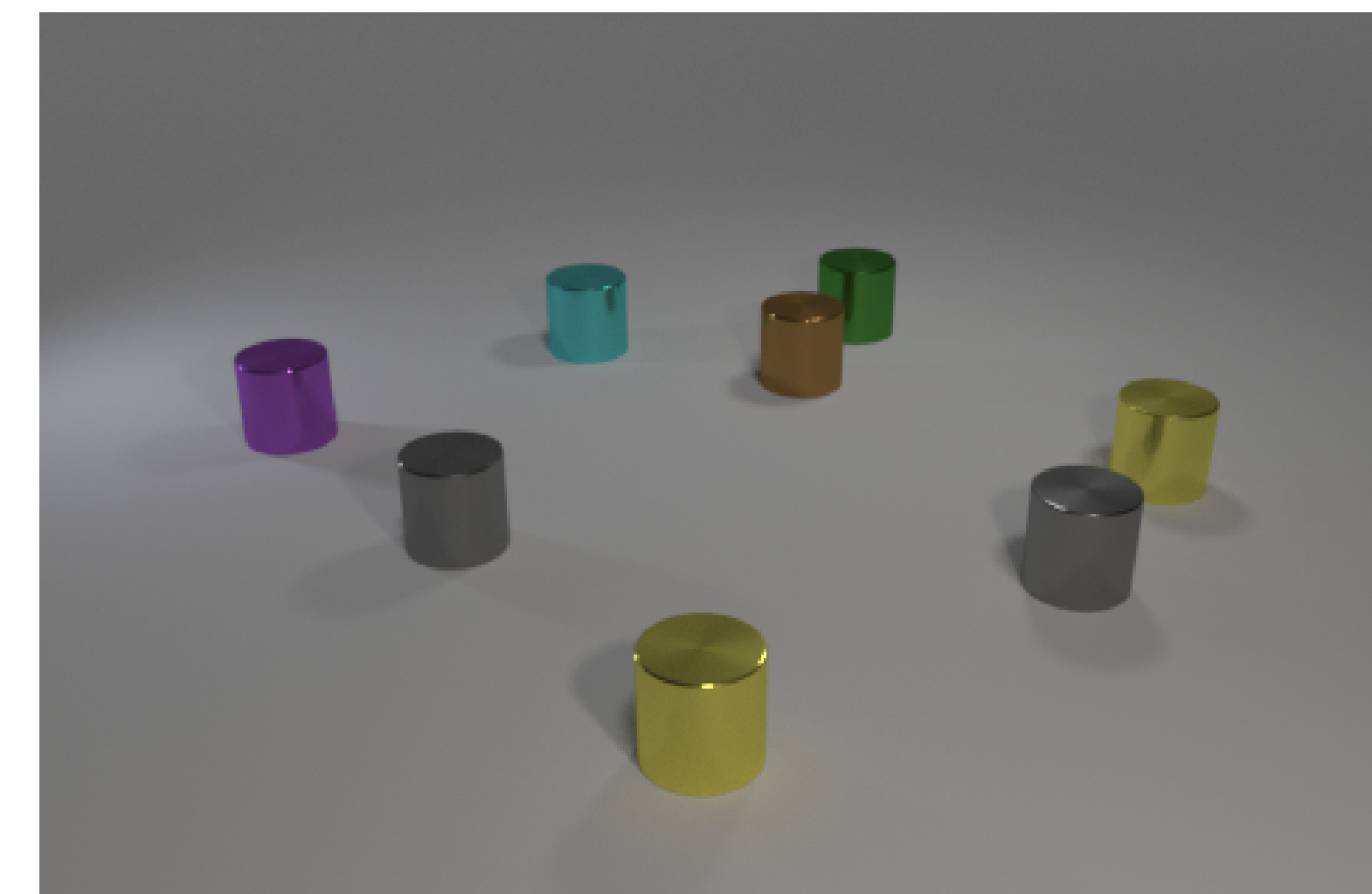


Learning curves by vocabulary size

RE Generation Results:

- **Accuracy:** 69%
- **F1-Score:** 89.53%
- **Shape:** 98.30% precision
- **Color:** 92.44% precision
- **Size:** 69.43% precision

CLEVR Color Dataset

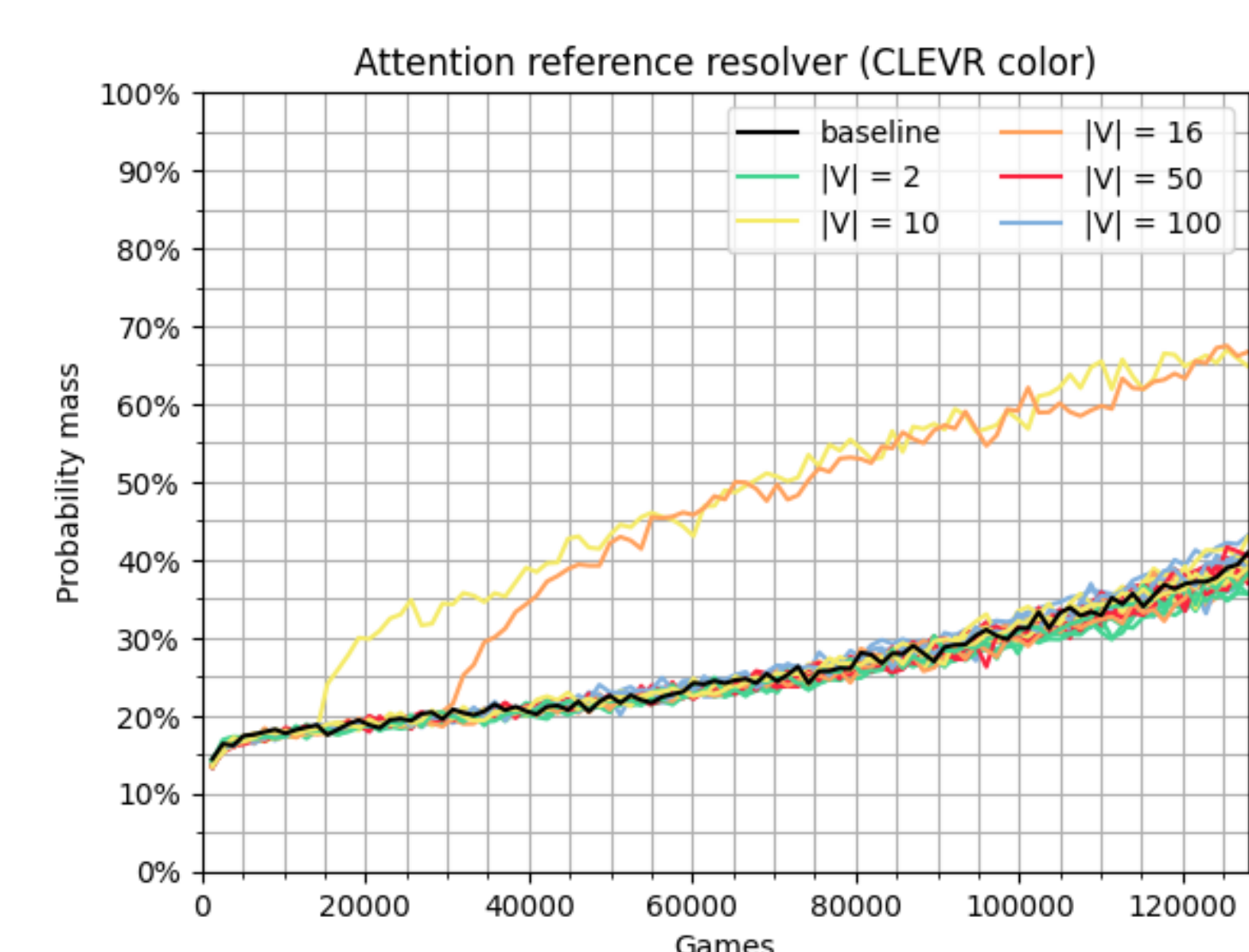


Target identified by color only

Sender in supervised generation:

- **strict string accuracy:** 93%
- **avg F1-Score on attributes:** 95.17%
- **Shape:** 100% precision
- **Color:** 92.76% precision
- **Size:** N/A (not used)

Sender and Receiver in interaction:



Learning curves by vocabulary size